

Chapter 20

Applications Of Inversive Geometry

From Japan With Love or Inversive Geometry Meets Sangaku

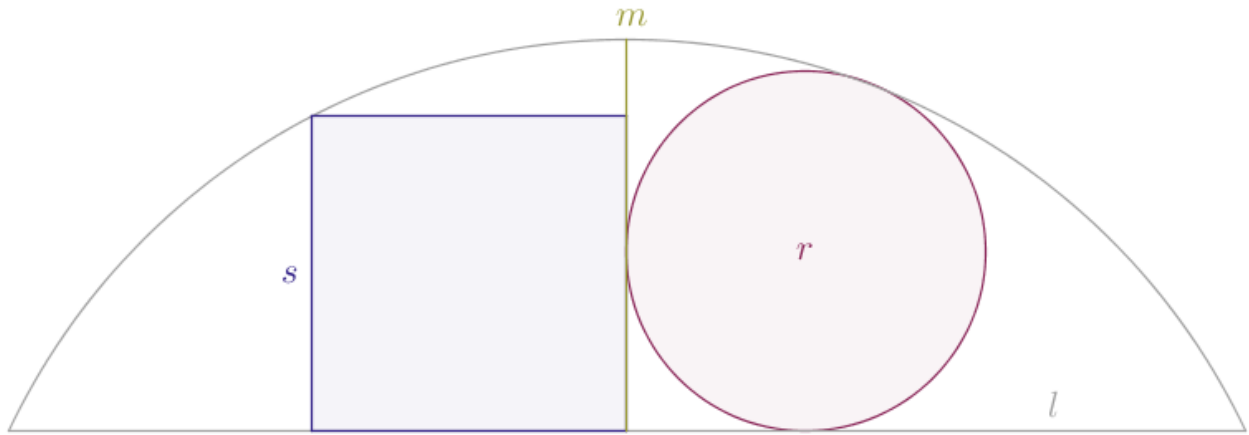
Explicitly or physically, the so-called *sangaku* were implemented as wooden plates painted with problems and theorems on plane and solid geometry generated by one or more experts in the field. Such wooden plates were normally hung by the entrances to the religious buildings so that other experts can see the problems and the theorems thus posted and, presumably, solve the problems or prove the theorems suggested.

Implicitly, *sangaku* were the means of the inter-expert communication in Japan during the 265-year Edo period, otherwise known as the *Great Peace*, that lasted between 1603 and 1868.

One characteristic feature of the physical nature of *sangaku* was the combination of an artistic elegance and exquisiteness of their rendering with the mathematical content.

Some sangaku problems are, apparently, so difficult that they are not solved to this day. Here is an example of one such problem.

A given circular segment, constructed on the chord of length l , is cut in half with a line segment of length m and into each half of the circular segment thus formed a square with the side length s and a circle of radius r is inscribed (Figure 20.1):



Define a real number a such that:

$$a = l + m + s + r$$

and a number b such that:

$$b = \frac{m}{l} + \frac{r}{m} + \frac{s}{r}$$

Objective: express the numbers l, m, s and r in terms of a and b .

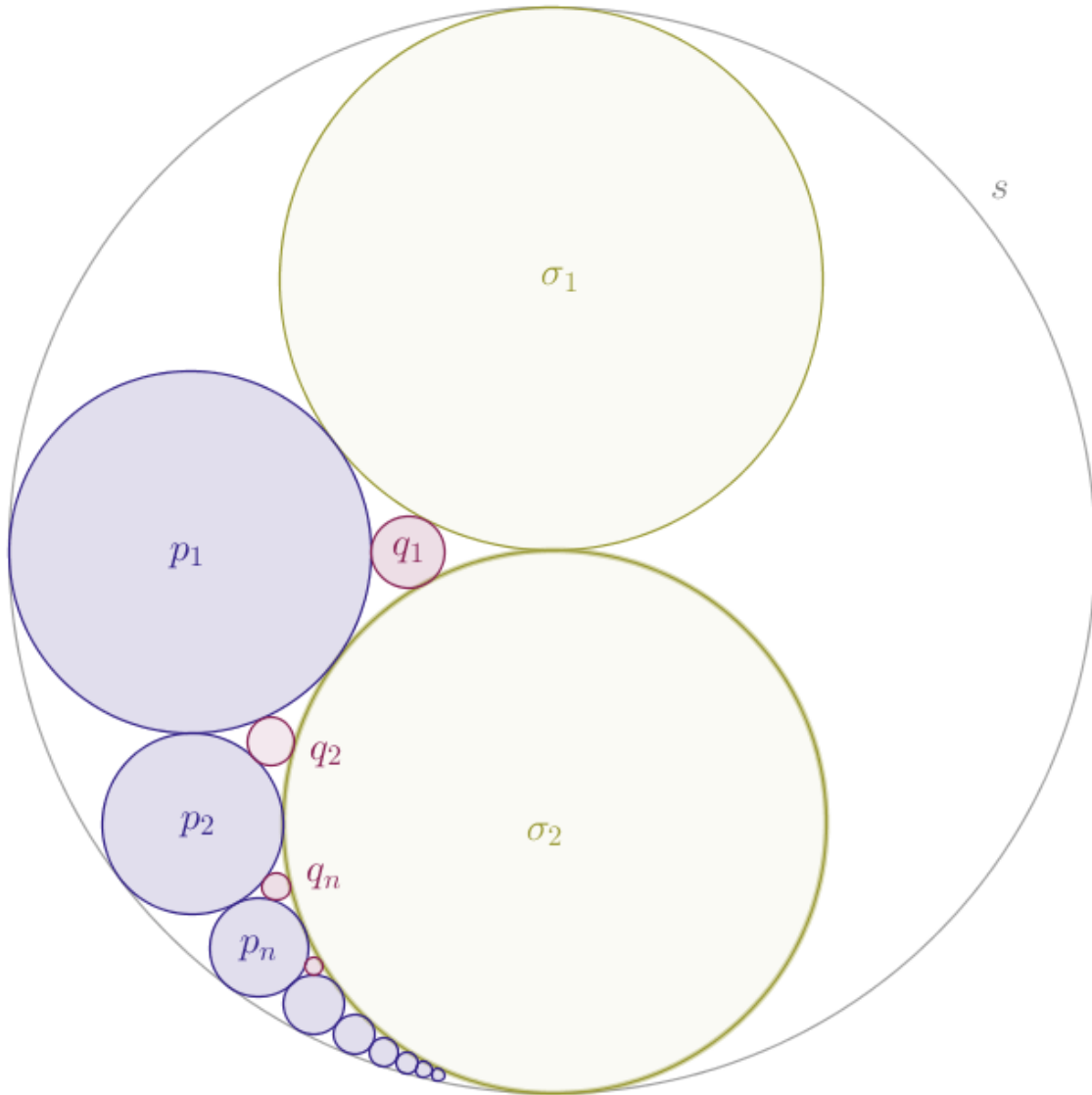
We, however, will use inversion to solve the following 1788 Hotta Jinsuke sangaku

Problem 26. Into a generating circle, s , of radius r two (green) circles, $\sigma_{1,2}$, of equal radius $r/2$ are inscribed in such a way that the circles $\sigma_{1,2}$ touch each other **externally**, while touching the generating circle, s , **internally**.

Into the curvilinear triangle thus formed a p -family of (blue) circles is inscribed as follows.

The first p -circle touches both circles $\sigma_{1,2}$ externally, while touching the generating circle, s , internally. For all subsequent whole positive numbers $n > 1$ it is the case that the n -th p -circle touches the $(n - 1)$ -st p -circle and the circle σ_2 externally, while touching the generating circle, s , internally.

Into the curvilinear triangles formed by the circles $\sigma_{1,2}$ and the (blue) p -circles an additional q -family of (red) circles is constructed as follows. The first q -circle touches the circles $\sigma_{1,2}$ and the first p -circle externally. For all subsequent whole positive numbers $n > 1$ it is the case that the n -th q -circle touches the circle σ_2 and both the $(n - 1)$ -st and the n -th p -circle externally (Figure 20.2):



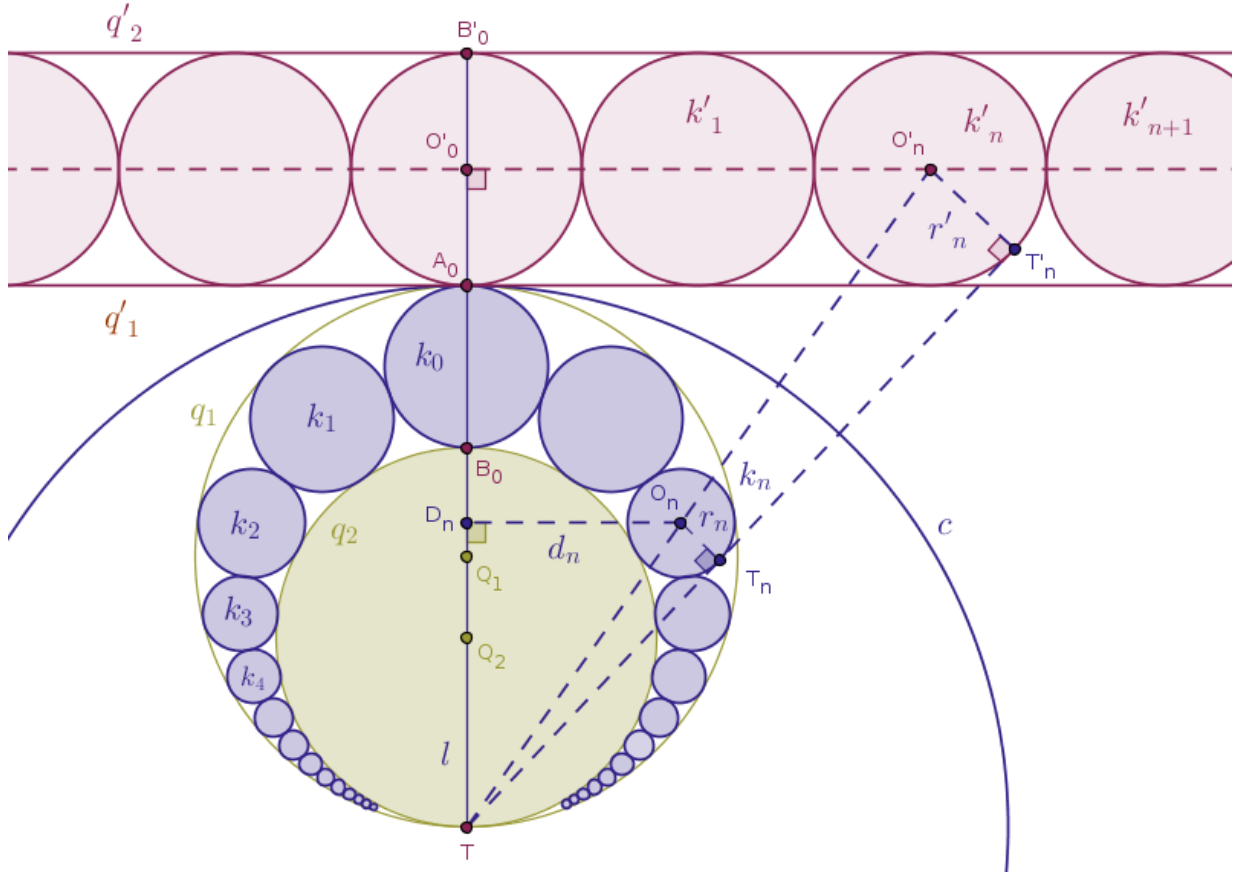
Let p_n denote the length of the radius of the n -th p -circle and let q_n denote the length of the radius of the n -th q -circle.

Objective: express the numbers p_n and q_n as the functions of n and r .

Solution. As usual, at first blush this problem seems to be grizzly and daunting.

However, those of our readers who solved on their own or at least worked their way through the **Chain of Pappus** problem from Chapter 12 already know that a transform-solve-reverse transform maneuver is coming up.

That is, if we recall our solution of the said problem (Figure 12.1):



where for the center of inversion we picked the point, T , where the two generating circles touch and where for the radius of inversion we picked the diameter of the larger circle, then in this particular and similar case it should be natural to investigate the inversion with positive power of the plane that houses our circles centered at the point, T , where the circles s and σ_2 touch.

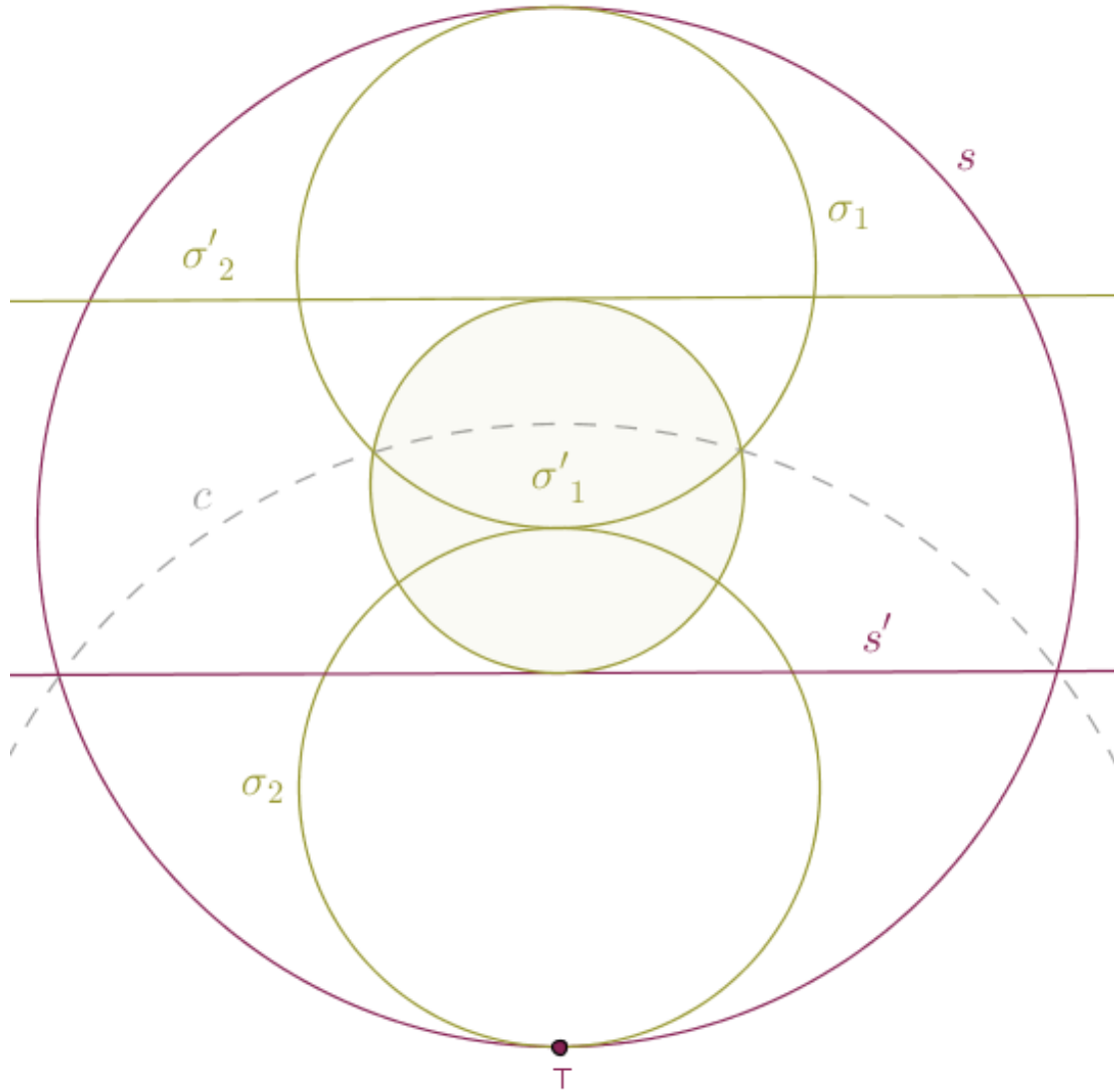
The concrete length of the radius, ρ , of our inversion here is irrelevant for we can choose it to be any reasonable (positive) value that our hearts desire.

It is one matter to be told to not worry about a certain aspect and it is another matter entirely to understand why the said aspect warrants no worrying about.

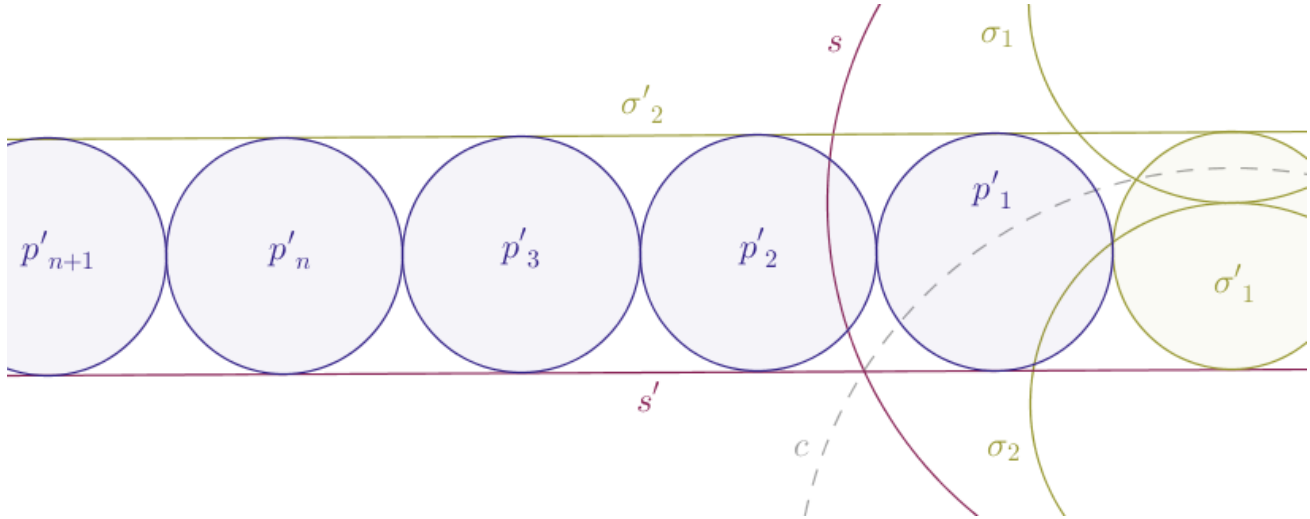
In order to show explicitly why we have made the above statement, we will carry out all the relevant computations assuming that the length of the radius of inversion is $\rho \neq 1$. By the end of our discussion, the readers should see for themselves why it was perfectly fine to put $\rho = 1$ right away.

Hence, under the inversion just described:

- the original circles s and σ_2 that are parallel will be transformed into the straight lines s' and σ'_2 that will be parallel due to the Theorems 5, 17, 19 and 35 and since both of these circles pass through the center of inversion
- the original circle σ_1 will be transformed into a circle σ'_1 trapped between or tangent to the straight lines s' and σ'_2 (Figure 20.3):



Observe how a successfully chosen inversion gives us a tactical advantage. Under such an inversion the images of the p -family of circles that are tangent to the the circles s and σ_2 will now be the circles that, being tangent to each other, are tangent to two straight lines that are parallel and, thus, the lengths of the radii of all such image p' -circles will be the same (Figure 20.4):

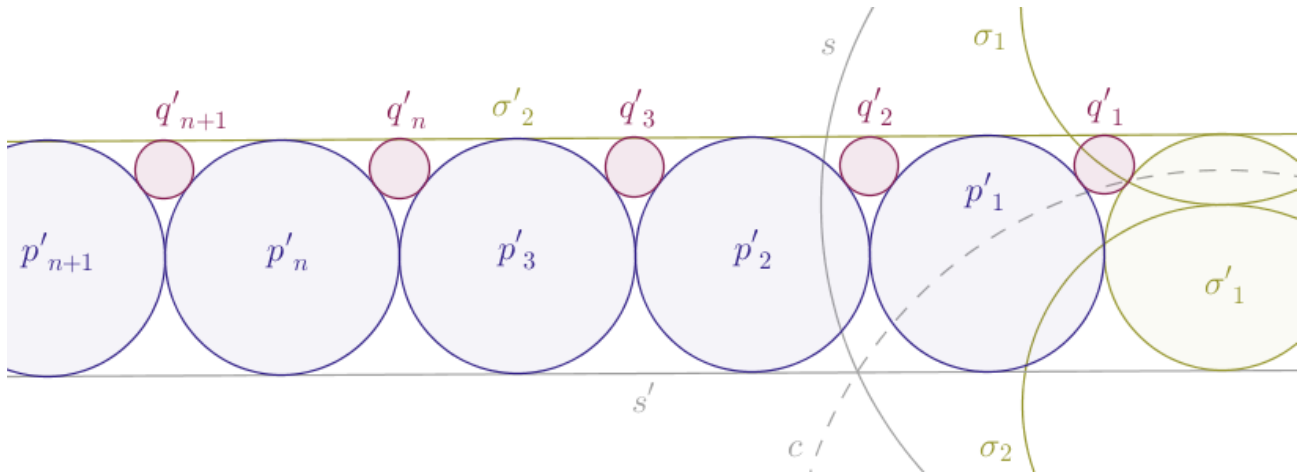


and by our earlier agreement, for the said radii p'_n we will have for any n :

$$p'_1 = p'_2 = p'_3 = \dots = p'_n = \dots = p' \quad (1)$$

Moreover, under such an inversion the images of the q -family of circles that are tangent to the the p -circles and the circle σ_2 will now be the circles each of which is tangent to the straight line σ'_2 and two tangent p' -circles of equal radius.

Thus, the lengths of the radii of all such image q' -circles will be the same as well (Figure 20.5):



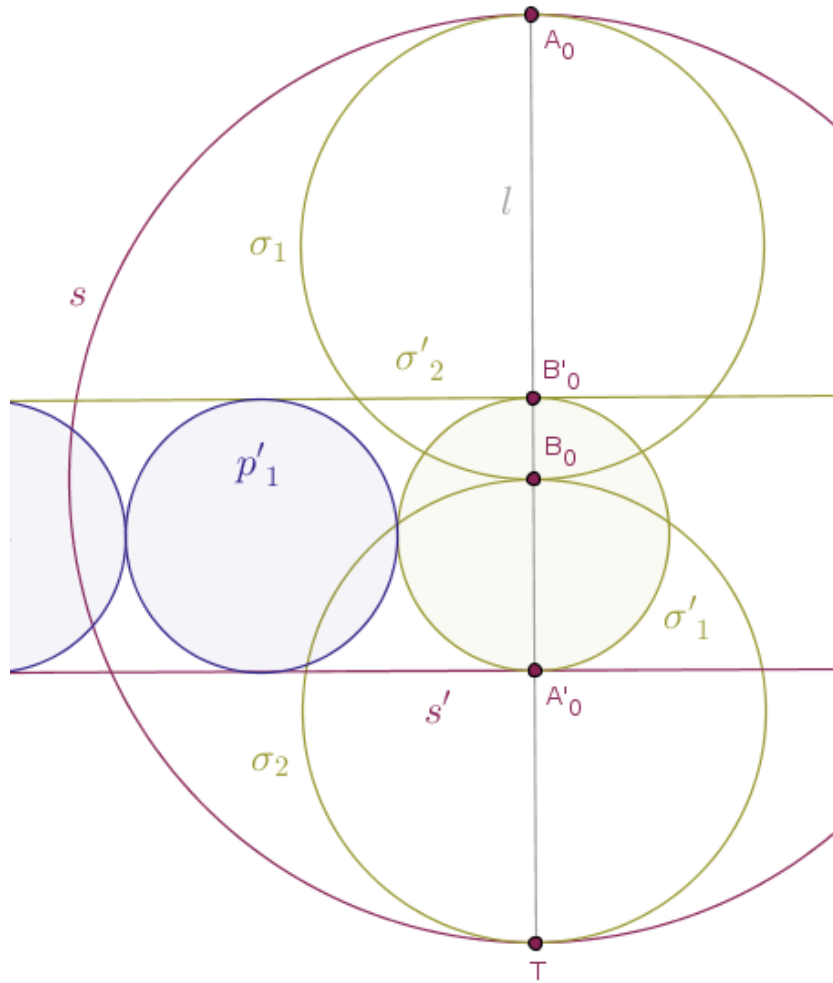
and by our earlier agreement, for the said radii q'_n we will have for any n :

$$q'_1 = q'_2 = q'_3 = \dots = q'_n = \dots = q' \quad (2)$$

Note that the task of inscribing a circle into a curvilinear triangle trapped between two tangent circles and a straight line tangent to the said circles has been solved by us in Chapter 17 already.

This is exactly how we generated all the diagrams for this problem and we already understand that by employing the inversive workhorse Condition 3 of Definition 1 and the Pythagorean theorem we expect to be able to get the handle on the task at hand.

In deed, by analogy with the solution of the Chain of Pappus problem, let l be the straight line that passes through the centers of the circles $\sigma_{1,2}$. Let l intersect the circles s and $\sigma_{1,2}$ at the points A_0 and B_0 correspondingly and let l intersect the straight lines s' and σ'_2 at the points A'_0 and B'_0 correspondingly (Figure 20.6):



But by our choice of inversion, the points A_0 and A'_0 are the inverses of each other.

Thus, by Condition 3 of Definition 1, for these two points we have:

$$|TA_0| \cdot |TA'_0| = 2r \cdot |TA'_0| = \rho^2$$

from where for the distance $|TA'_0|$ we find:

$$|TA'_0| = \rho^2 \cdot \frac{1}{2r} \quad (3)$$

Likewise, under our choice of inversion, the points B_0 and B'_0 are the inverses of each other:

$$|TB_0| \cdot |TB'_0| = r \cdot |TB'_0| = \rho^2$$

from where:

$$|TB'_0| = \rho^2 \cdot \frac{1}{r} \quad (4)$$

But, clearly, the radius p' of each p' -circle is equal to the radius of the circle σ'_1 , which is just the distance between the points A'_0 and B'_0 halved:

$$p' = \frac{|TB'_0| - |TA'_0|}{2} = \rho^2 \cdot \frac{1}{4r} \quad (5)$$

Now that we trivially solved the inverted problem, we would like to find an algebraic relationship between the numbers p' and p .

To that end, we mimic the idea that we already uncovered during the solution of the Chain of Pappus problem.

Namely, we draw a tangent straight line TT'_n to two n -th p - and p' -circles through the point T and we then consider two similar right triangles TT_nP_n and $TT'_nP'_n$, where the points P_n and P'_n are the centers of the said circles.

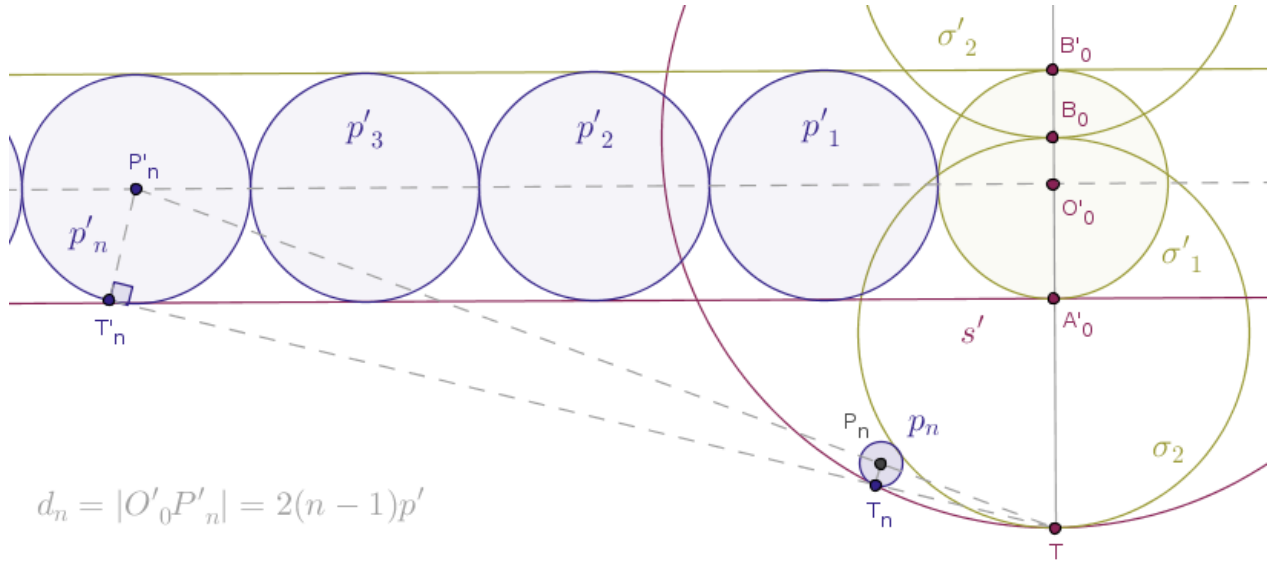
Since the triangles TT_nP_n and $TT'_nP'_n$ are similar, because they are both right and both share a common interior angle at the vertex T , it follows that the lengths of their corresponding sides are in the same proportion:

$$\frac{p_n}{p'} = \frac{|TT_n|}{|TT'_n|}$$

from where:

$$p_n = p' \cdot \frac{|TT_n|}{|TT'_n|}$$

But the tangency points T_n and T'_n (Figure 20.7):



are the images of each other under the chosen inversion:

$$|TT_n| \cdot |TT'_n| = \rho^2$$

Hence:

$$|TT_n| = \frac{\rho^2}{|TT'_n|}$$

and, hence, for the length p_n we have:

$$p_n = p' \cdot \frac{\rho^2}{|TT'_n|^2} \quad (6)$$

The square of the distance $|TT'_n|$ plays the role of the power of the point T with respect to the n -th p' -circle:

$$|TT'_n|^2 = |TP'_n|^2 - p'^2$$

The distance $|TP'_n|$, however, can be recovered from the right triangle $TO'_0P'_n$:

$$|TP'_n|^2 = |TO'_0|^2 + |O'_0P'_n|^2$$

Our analog of the distance $d_n = |O'_0P'_n|$ from the Chain of Pappus problem is offset by 1 because we number our circles starting from 1 and not from 0:

$$d_n = |O'_0P'_n| = 2(n-1)p' \quad (7)$$

and the distance $|TO'_0|$ is also known to us, since we already computed the length p' in (5):

$$|TO'_0| = |TA'_0| + |A'_0O'_0| = \rho^2 \cdot \frac{1}{2r} + \rho^2 \cdot \frac{1}{4r}$$

which is to say that:

$$|TO'_0| = \rho^2 \cdot \frac{3}{4r} \quad (8)$$

Thus:

$$|TP'_n|^2 = \rho^4 \cdot \frac{9}{16r^2} + 4(n-1)^2 p'^2$$

and:

$$|TT'_n|^2 = \rho^4 \cdot \frac{9}{16r^2} + 4(n-1)^2 p'^2 - p'^2 \quad (9)$$

Taking the value of p' from (5) and putting it into (9), we find:

$$|TT'_n|^2 = \rho^4 \cdot \frac{1}{2r^2} + \rho^4 \cdot \frac{(n-1)^2}{4r^2} \quad (10)$$

Putting the RHS of (10) back into (6), we arrive at the expression that captures the length p_n as a function of n and r :

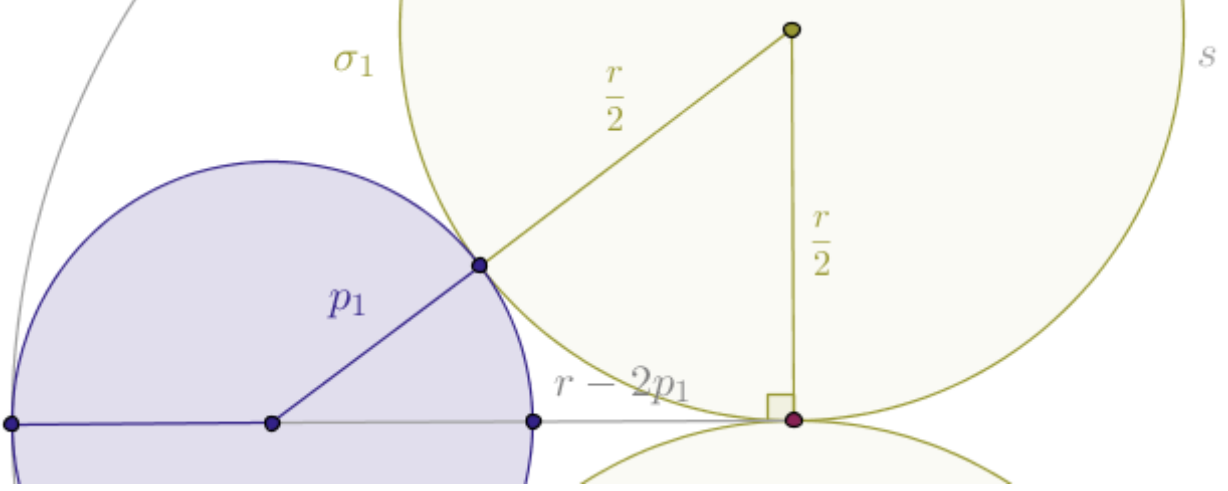
$$p_n = \rho^2 \cdot \frac{1}{4r} \cdot \frac{\rho^2 \cdot 4r^2}{2\rho^4 + (n-1)^2 \rho^4}$$

and we now understand why we stated earlier that the actual length of the radius of inversion ρ in this case is irrelevant.

Because in the expression above all the copies of ρ^4 cancel out:

$$p_n(n, r) = \frac{r}{2 + (n - 1)^2}, \forall n > 1 \quad (11)$$

Do note that the length p_1 of the radius of the first p -circle when $n = 1$ can be computed trivially via the Pythagorean theorem (Figure 20.8):



from the right triangle shown above:

$$\left(p_1 + \frac{r}{2}\right)^2 = \frac{r^2}{4} + (r - p_1)^2$$

Thus, for p_1 we have:

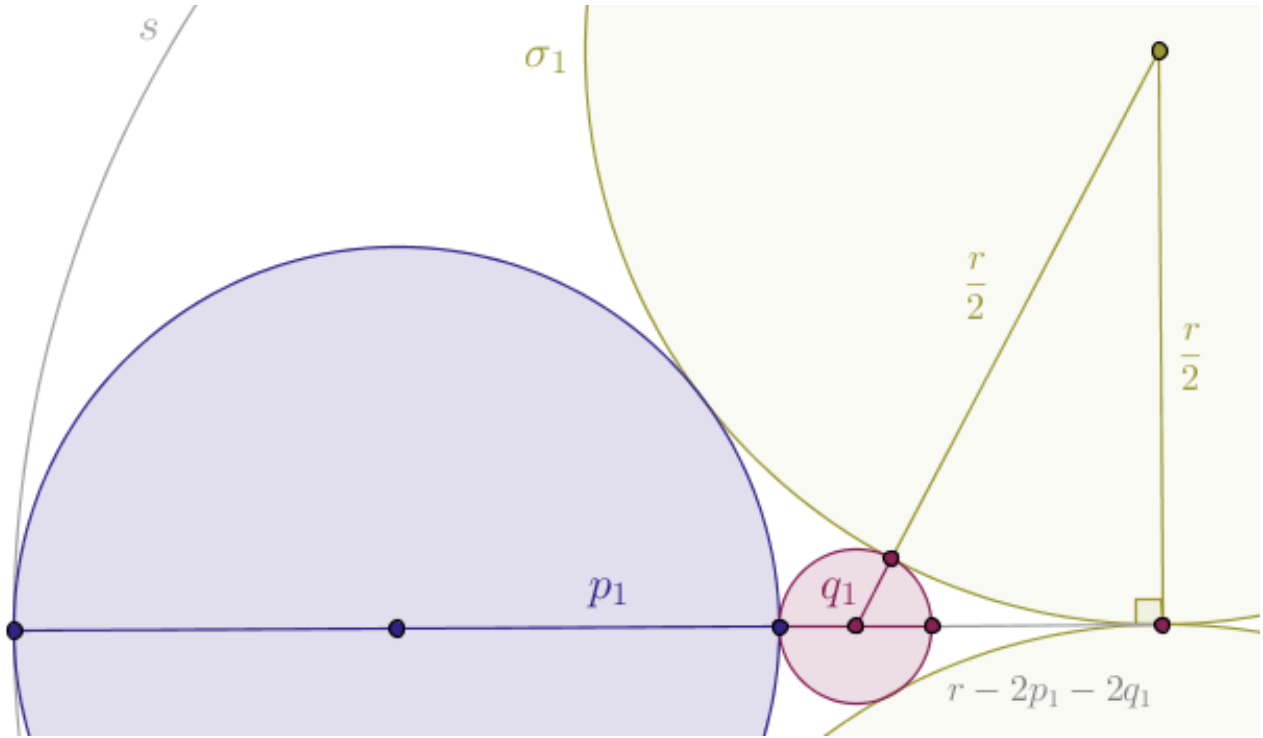
$$p_1 = \frac{r}{3}, n = 1 \quad (12)$$

and now that we have solved the given problem for the blue family of p -circles completely, we move onto the solution of this problem for the red family of q -circles.

The Red Family Of q -Circles

Here we recover the length of the radius q_1 of the first q -circle, when $n = 1$, in exactly the same way as we just did it for the first blue p -circle, p_1 , using the Pythagorean theorem.

From the right triangle shown below (Figure 20.9):



we have:

$$\left(q_1 + \frac{r}{2}\right)^2 = \frac{r^2}{4} + (r - 2p_1 - q_1)^2$$

Thus, with the value of p_1 from (12), we find q_1 for $n = 1$:

$$\boxed{q_1 = \frac{r}{15}, n = 1} \quad (13)$$

and we now can determine the length of the radius q' of all the q' -circles directly via the handy Theorem 31 which we proved at the end of Chapter 7 and which establishes an algebraic relationship between the radii of a preimage and the image circles under any inversion of the real plane with respect to a circle and which is applicable for any power of inversion, positive and negative.

Namely, if R and R' are the lengths of the radii of a preimage and the image circles correspondingly and d is the distance from the center of inversion to the center of the preimage circle then:

$$R' = R \cdot \frac{r^2}{|d^2 - R^2|}$$

if we agree that r is the radius of the circle of inversion.

Thus, according to Theorem 31 in our case, for the length q' of the radius of the first inverted q' -circle we have:

$$q' = q_1 \cdot \frac{\rho^2}{|d^2 - q_1^2|}$$

where the distance $d = |TQ_1|$ can be readily recovered from the right triangle Q_1B_0T (Figure 20.10):



via the Pythagorean theorem:

$$d^2 = |TQ_1|^2 = r^2 + (r - 2p_1 - q_1)^2$$

which, with (12) and (13), yields:

$$d^2 = |TQ_1|^2 = r^2 + \frac{4^2 r^2}{15^2}$$

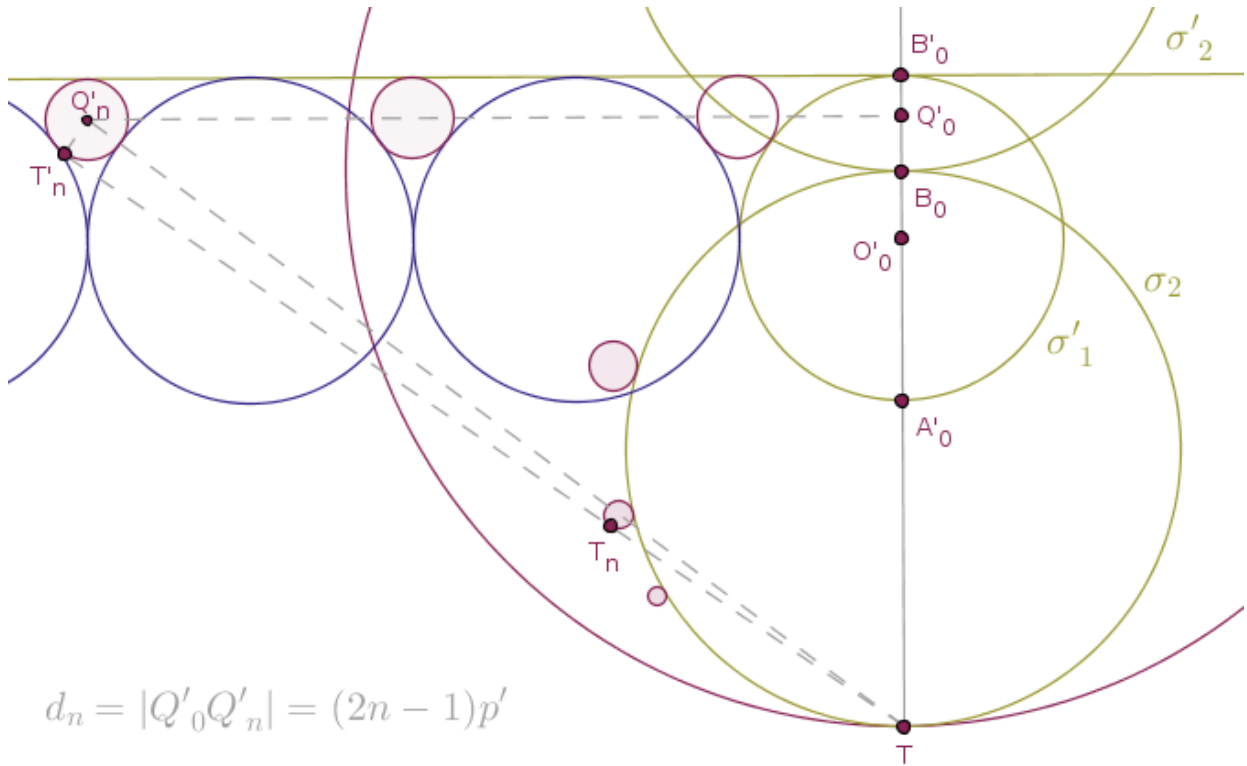
Thus:

$$q' = q_1 \cdot \frac{\rho^2}{r^2 + \frac{4^2 r^2}{15^2} - q_1^2} = \rho^2 \cdot \frac{r}{15} \cdot \frac{1}{r^2 + \frac{4^2 r^2}{15^2} - \frac{r^2}{15^2}} = \rho^2 \cdot \frac{r}{15} \cdot \frac{1}{r^2 + \frac{r^2}{15}}$$

which is to say that:

$$q = \rho^2 \cdot \frac{1}{16r} \quad (14)$$

and we are almost done (Figure 20.11):



Applying exactly the same logic as we used above with the p -family of circles to the q -family of circles, we replay exactly the same steps of constructing a common exterior tangent straight line to the n -th q and q' -circles through the point T as shown in Figure 20.11 above.

This time we, by analogy, will have:

$$q_n = q' \cdot \frac{\rho^2}{|TT'_n|^2}$$

and for the power of the point T with respect to the n -th q' -circle we will have:

$$|TT'_n|^2 = |TQ'_n|^2 - q'^2$$

where the square of the distance $|TQ'_n|$ is recovered from the right $TQ'_0Q'_n$ triangle:

$$|TQ'_n|^2 = |TQ'_0|^2 + |Q'_0Q'_n|^2$$

Because horizontally the center Q'_n of the n -th q' -circle sits exactly between the centers of the adjacent p' -circles, this time for the distance $d_n = |Q'_0Q'_n|$ we will have:

$$d_n = |Q'_0Q'_n| = (2n - 1)p'$$

and for the distance $|TQ'_0|$ we trivially have via (4):

$$|TQ'_0| = |TB'_0| - q' = \rho^2 \cdot \frac{15}{16r}$$

Hence:

$$|TQ'_n|^2 = \rho^4 \cdot \frac{15^2}{16^2 r^2} + (2n - 1)^2 p'^2$$

and:

$$|TT'_n|^2 = \rho^4 \cdot \frac{15^2}{16^2 r^2} + (2n - 1)^2 p'^2 - q'^2$$

which, with (5) and (14), is to say that:

$$|TT'_n|^2 = \rho^4 \cdot \frac{7}{8r^2} + \rho^4 \cdot \frac{(2n - 1)^2}{16r^2}$$

since:

$$15^2 - 1 = (15 - 1)(15 + 1) = 14 \cdot 16$$

Putting all of the above intermediate results together, we find for q_n :

$$q_n = \rho^2 \cdot \frac{1}{16r} \cdot \frac{\rho^2}{\rho^4 \cdot \frac{7}{8r^2} + \rho^4 \cdot \frac{(2n-1)^2}{16r^2}}$$

noting that, again, all the copies of ρ^4 cancel out:

$$\boxed{q_n(n, r) = \frac{r}{14 + (2n - 1)^2}, \forall n > 1} \quad (15)$$

and the problem is now solved completely.

We see that the ratio of p_n to q_n does not depend on r :

$$\boxed{\frac{p_n}{q_n} = \frac{14 + (2n - 1)^2}{2 + (n - 10)^2}} \quad (16)$$

and that we can also solve the relations (11) and (15) for n :

$$n = 1 + \sqrt{\frac{r}{p_n} - 2} \quad (17)$$

and

$$n = \frac{1}{2} \cdot \left(1 + \sqrt{\frac{r}{q_n} - 14} \right) \quad (18)$$

As the above solution clearly demonstrates, this particular sangaku problem bears a fair amount of resemblance to the Chain of Pappus problem. $\square$